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Soil-water-conservation strategies for sustainable agriculture in changing climate

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ABSTRACT

Climate change-induced intense hydrological cycles and extreme rainfall events are potentially accelerating the erosion and sediment-yield rates by changing the rainfall–runoff erosivity. The accelerated runoff and soil loss cause loss of the top fertile soil and organic carbon, reduces nutrient and soil-water availability, thus results in significant loss in crop productivity. Hence, soil-erosion control is a pre-requisite to achieve the agricultural sustainability in a climate-change scenario. Various soil-management strategies have potential to achieve the agricultural sustainability under climate-change scenario. Among the conservation technologies, agronomic measures such as proper tillage and land configuration, compartmental bunding, broad-bed furrow, ridge and furrow systems, broad-bed furrow system and conservation furrows have proved to be highly effective in arable lands and significantly reduce the impact of extreme weather events on natural soil resources. Other agronomic practices such as mulching, conservation agriculture, cover crops and vegetative barrier have also proved highly effective against erosion and improve crop productivity besides maintaining the soil health. Water-harvesting structures and recycling of excess runoff water potentially reduce the impact of climate change on the natural resources. Use of geotextiles protect the soil from rainfall drops, and have major role in control of river-bank erosion and slope stabilization in high-rainfall regions. Practice of agroforestry systems in degraded ecosystems improves soil quality and enhances soil productivity through biological nitrogen fixation, soil-carbon sequestration, efficient nutrient cycling, and deep capture of nutrients and water from soils.

Key words: Broad bed and furrow, Compartmental bunding, Degraded ecosystems, Geo-textiles, Water harvesting

INTRODUCTION

The status of global soil resources published by the FAO (2015) highlights that, 'the majority of the world's soil resources are in only fair, poor, or very poor condition'. Soil erosion is still a major threat to ecosystem stability and agricultural productivity, worldwide. More-intense hydrological cycles and extreme-rainfall events induced by the changing climate are potentially accelerating the erosion rates. Climate change is most likely to increase the soil erosion and sediment-yield rate by changing the rainfall–runoff erosivity. The extreme rainfall events and continuously changing precipitation patterns are accelerating the rainfall–runoff erosivity and thereby impacts the soil-erosion process (Talchabhadel *et al.,* 2020). Accelerated soil erosion as a result of change in the rainfall pattern also depletes the soil organic carbon. Clay loss and organic carbon loss were 4.9 t/ha and 185 kg/ha at 28% slope in Nilgiri hills and these losses resulted in severe productivity loss in cabbage (*Brassica aleracea* var. capitata L.) crop (Kannan *et al.,* 2019). Mondal *et al.* (2016) quantified the impact of climate change on future soil erosion and soil organic carbon under different slope and land use categories in Narmada river basin in India and reported that sediment load changed by 5.33, 17.97 and 58.37% in the 2020s, 2050s and 2080s, respectively from the current erosion rate.

Many studies have clearly indicated significant loss in crop productivity due to the soil erosion. Moreover, loss of the fertile top soil can have considerable impact on nutrient availability; soil-water availability and plant-growth properties. In a predominantly agricultural system, the objective of improving the productivity, profitability and prosperity of the farmers and achieving agricultural development on an ecologically sustainable basis can be attained only when conservation of the natural resources is assured. Soil erosion can be mitigated using sustainable land-management techniques based on the engineering, agronomic, biological and scientific land-management

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practices. This review paper addresses various soil-management strategies, to achieve the agricultural sustainability under climate change scenario. The losses caused by extreme-climatic events like high-intensity rainfall, drought, degradation of soil quality can be minimized if appropriate soil- and water-conservation measures are adopted. In rainfed farming, information on rainfall pattern and moisture deficits over time is very helpful in crop planning, rainwater management and other hydrological studies related to agriculture.

AGRONOMIC MEASURES FOR SOIL AND WATER CONSERVATION

Biological or vegetative measures are preferred in soil- and water-conservation programmes, as they are eco-friendly, sustainable and cost-effective (Sharda *et al.,* 2006). These can be adopted alone or in combination with mechanical measures, depending on the intensity of soil erosion on moderate land slope.

Tillage

Land and water are closely interconnected and consequently they influence land productivity, therefore, land-management techniques that encourage more rainfall to enter the soil are key strategies for improving productivity of rainfed systems. Tillage roughens the soil surface and breaks any soil crust. This leads to increased water storage by increased infiltration (Ali and Talukder, 2008). However, response to tillage varies with rainfall, soil type and kind of crops. Crops like maize (*Zea mays* L.), pigeonpea (*Cajanus cajan* (L.) Millsp.), cotton (*Gossypium hirsutum* L.), castor (*Ricinus communis* L.), soybean [*Glycine max* (L) Merr.] and sunflower (*Helianthus annuus* L.), respond very favourably to deep tillage. Patil *et al.* (2016) reported that conventional tillage conserved greater rainfall and improved soil physical properties, grain yield and water use efficiency (WUE) compared to the reduced tillage in Vertisols of Karnataka (table 1).

Sharda *et al.* (2006) reported that, deep tillage has a definite edge over shallow tillage in improving the yields of different dryland crops in red soils of southern India irrespective of type of season (Table 2). Deep tillage with plough followed by chiseling/ cultivator (Channappa, 1994) opens the hard layers and increases the infiltration rate, water-storage capacity and greater availability of soil moisture to plants during drought in red soils at Bengaluru, Karnataka and Coimbatore, Tami Nadu.

Land configuration

Management of rainwater, especially *in-situ* conservation, is an important component of resource-conservation practices for augmenting crop productivity in dryland areas. Few of the land configurations, viz. contour farming,

Table 1. Soil moisture, infiltration, sorghum yield and water use efficiency (WUE) as influenced by tillage

Tillage	Soil moisture (mm/60 cm depth) at the time of sowing	Infiltration (cm/hr)	Sorghum grain yield (kg/ha)	WUE (kg/ha/mm)
Conventional tillage	278	9.0	433	2.21
Reduced tillage	210	8.2	388	2.0
Less tillage	195	7.4	343	1.87

Source: Patil *et al.* (2016)

Table 2. Effects of tillage on yield of different crops in Southern India

Type of season	Crop	Grain yield (t/ha)		Rainfall during the crop-growth period (mm)
		Shallow tillage	Deep tillage	
Sub normal	Sorghum	1.74	1.81	264
	Finger millet	0.10	0.15	124
	Castor	0.38	0.43	264
Normal	Pigeonpea	0.25	0.29	149
	Sorghum	2.66	2.95	259
	Pearl millet	1.94	2.23	399
Above normal	Castor	0.08	0.13	621
	Sorghum	2.61	2.65	609
	Pearl millet	1.60	1.63	529
	Castor	0.96	1.27	883

Source: Sharda *et al.* (2006)

Table 3. Benefits of contour farming

Activity/ soil	Benefit	Reference
Contour farming with vegetative strips in Alfisols	Reduction in sediment yield by 63%	Mourad and Joseph (2018)
Contour farming in clay soils	10% reduction in runoff and 49% reduction in soil loss	Farahani *et al.* (2016)
Contour farming with intercrop in rainfed Alfisols	Higher soil moisture and maize equivalent yield	Ramanjaneyalu *et al.* (2020)
Contour farming in Alfisols	25 and 28% higher yield in finger millet and groundnut, respectively	Krishnappa *et al.* (1994)
Contour farming in Vertisols	68% to 85% higher yield in winter sorghum across varies slopes	Ramamohanrao *et al.* (2000)
Contour farming and intercrop + greengram	Increased yield of crops by 15.4 to 26% and reduced soil loss by 17.6 to 25%	Sharda *et al.* (2006)
Ridges and furrows on contour	Increased infiltration and potato equivalent yield by 8%, reduced soil loss by 30%	Kannan and Madhu (2013)

land smoothening, dead furrows, compartmental bund, broad bed and furrow and raised- and sunken-bed system are very effective *in-situ* rainwater-conservation particularly in low-rainfall areas.

Contour cultivation or cultivation across slope: Carrying out all the field operations including sowing of crops across the slope and along contour (contour cultivation) provides a series of miniature barriers to running rainwater and reduces runoff, soil loss and increases soil water and nutrient storage in soil profile (Table 3). Contour cultivation is recommended for all types of soils, rainfall up to 1,000 mm and slope varying from 0.5 to 4%. It helps in reduction of runoff by impounding rainwater in small depressions and reduces the development of rills. The effectiveness of this practice varies with rainfall, soil type and topography. Maximum effectiveness of this practice is on medium slopes and on permeable soil. The relative effectiveness decreases as the land grade becomes very flat or very steep. On long slopes, where bunding is done to decrease the slope length, the bunds can act as guidelines for contour cultivation. Contour cultivation on steep slopes or under conditions of high-rainfall intensity may cause formation of gullies because row breaks may release the stored runoff water to next downstream row.

Compartmental bunding: These are practices in medium and black soil area to store rainwater in the soil profile during the monsoon for the use of winter (*rabi)* crop. Compartmental bunds provide greater opportunity time for rainwater to infiltrate into the soil and wet the soil profile completely for early sowing of winter crops, thus giving greater crop yields. The size of the compartmental bunding varies with slope of the field. Compartments of 6 m × 6 m up to 1% slope; 4.5 m × 4.5 m for 1–2% slope, and 3 m × 3 m for 2% slopes are recommended (Sharda *et al.,* 2006). Effect of compartmental bunding on crop productivity was more during a drought year compared to normal rainfall situation. Increased soil-moisture content and 36.7% higher chickpea (*Cicer arietnum* L.) yield owing to compartmental bunding was also reported by Patil *et al.* (2016) in Vertisols of Ballary area. Pandian *et al*. (2020) also reported the enhancement in yield of groundnut (*Arachis hypogaea* L.), water-use efficiency and organic carbon storage in Alfisols.

Ridges and furrows: Formation of ridges and furrows has been found most suitable for soil-moisture conservation and to reduce runoff and soil loss, particularly in light soils. Cultivation of crops under ridge-and furrow-system across the major land slope, with a gradient of 0.2–0.4% in land, having 1–3% slope will conserve more rainwater *in-situ*. This is suitable for widely spaced crops with 60 cm or more row spacing. A field length of 60 to 90 m is optimum for cultivation of crops with ridges and furrows. Lal *et al.* (1984) found that, the adoption of a 'W'-form *in-situ* rainwater-harvesting system demonstrated effective results on crop production in Brazil. In Vertisols of Bellary, Karnataka, ridges and furrows were more effective in conservation of rainwater and increased more grain yield of winter sorghum [*Sorghum bicolor* (L.) Moench] during a drought year (36%) compared to normal year (16%).

Broad-bed furrow (BBF) system: The BBF developed by the International Crop Research Institute for the Semi-Arid Tropics, Hyderabad, (ICRISAT), India, for increasing the productivity of semi-arid, poorly drained Vertisols, provides more opportunity for infiltration of rainwater and at the same time prevents waterlogging of the crop growing on the bed. The BBF system consists of a relatively raised-flat bed or ridge approximately 95 cm wide and shallow furrow about 55 cm wide and 15 cm deep across the slope on a grade of 0.2–0.6% for optimum performance. The bed width also depends on the crops, soil type and rainfall. The furrow acts as drainage for removing excess water and broad bed stores rainwater. In black soil,

Table 4. Effect of broad and furrow system on groundnut yield and soil loss

Treatment	Runoff (%)	Soil loss (kg/ha)	Soil moisture (%)	Groundnut yield (kg/ha)	Benefit: Cost ratio
Flat bed 45 cm row spacing	25.6	582	24.24	832	2.5
Broad bed (90 cm) and furrow (30 cm) with 3 crop rows	22.7	483	25.87	932	2.7

Source: Vekariya *et al*. (2015)

crops are sown in pre-formed beds made before the season and maintained year after year. This is suitable for narrow-spaced row crops. Even if a few rows are lost due to the furrow, the yields are made up owing to better *in-situ* rainwater conservation. Vekariya *et al.* (2015) found that, adoption of broad bed (90 cm width) and furrow (45 cm) with 3 rows was found superior for groundnut yield, net returns along with benefit: cost (B: C) ratio for maintaining higher soil moisture in root zone as well as for minimizing runoff and soil loss in medium black soils under rainfed condition of North Saurashtra Agro-climatic Zone of Gujarat (Table 4).

Conservation furrow system: The conservation furrow is a simple and low-cost *in-situ* soil-and rainwater-conservation practice adopted in Alfisols and associated soils with problems of crusting and sealing for rainfed areas (400–900 mm rainfall) with moderate slope varying from 1 to 4%. Due to crusting, early runoff is quite common in these soils. Furrows at 3–5 m apart on contour or across slope are opened either during planting or during intercultural operation using country plough. These furrows harvest the local runoff water and improve the soil moisture in the adjoining crop rows, particularly during the period of water stress. The practice has been found to increase the crop yields by 10–25%. Tewodros (2009) reported that, conservation furrow decreased runoff by 42% and soil loss by 62% in Ethiopia at moderate slopes.

Conservation bench terracing (CBT) or zing terracing: This type of terracing is adopted in 3-10% land slope in medium to deep soils with high-rainfall areas. The aim of constructing this kind of terraces is to cut down the length of slope, and to harvest the runoff from upper areas for the benefits of crops grown in lower areas. The upper 50 to 75% of the field is used for cultivating rainfed crops and lower 25–50% is used for cultivating water-demanding crops like rice (*Oryza sativa* L.) from the harvested water. Usually water-intensive crops are cultivated in the levelled portion (receiving area), while dry crops are cultivated in the unlevelled (donor) area. In the levelled one-third portions, normal crop can be harvested even during severe drought year and it is possible to cultivate 2 crops during a normal year. This will increase both cropping intensity and crop yields in the region. In Vertisols of Bijapur, Karnataka, layout of field with zing terrace increased the winter sorghum and safflower (*Carthamus tinctorius* L.) yields by 4 and 30%, respectively, over the control (AICRPDA, 1989; 1990). The effect of zing terrace was more felt in the levelled portion than the unlevelled portion. In the levelled portion, the yields of winter sorghum and safflower increased by 25 and 44%, respectively, over the control.

Mulching and residue management

Organic mulching: Mulches dissipate the kinetic energy of the rain drops, prevent soil erosion (splash erosion), facilitate infiltration, regulate soil temperature, improve the water-holding capacity of the soil. As a result, supplemental water demand of the crops is reduced. Mulching is a useful practice in controlling erosion, weed growth and conserve moisture as well as nutrient in the soil in rainfed hilly region (Sharma *et al.,* 2010a). Application of mulch was found to influence the soil physical characters positively. Chakraborty *et al.* (2010) reported that, organic mulching using rice husk increased the soil moisture (3%), produced more roots (25 and 40% higher root weight and root-length density), increased wheat grain yield (13–21%) and water-use efficiency (25%) in semi-arid environment. More importantly, mulching improves the burrowing activities of earthworms and improves air-moisture balance in the soil. Besides improving soil physical properties, like better drainage in clayey soil, mulches improve soil micro-nutrients and microbial population.

Vertical and live mulching: Vertical mulching involves opening trenches of 30 cm depth and 15 cm width across the slope at vertical intervals of 30 cm and stuffing sorghum stubbles vertically in these trenches, so that they protrude 10 cm above the ground. Vertical mulches of sorghum act as intake points and guide runoff water into subsoil layers thus, increase profile soil moisture and increased winter sorghum yields to a greater extent in a dry/ drought year compared to wet/ normal or above-normal rainfall years. This technique in medium to deep stiff and clayey soils enhanced sorghum yield varying from 26 to 78%, respectively, (Ramamohanrao *et al.,* 1978).

Live mulch is a cover crop, preferably leguminous crop

Table 5. Effect of legume mulching on soil properties and yield of maize and wheat

Treatment	Bulk density (Mg/m^3)	Infiltration rate (mm/hr)	Organic C (%)	Maize yield (t/ha)	Wheat yield (t/ha)
Control	1.44	6.33	0.57	2.05	1.86
Sunhemp	1.41	7.36	0.64	2.23	2.04
Leucaena	1.36	7.67	0.66	2.19	2.06
Sunhemp + leucaena	1.36	8.34	0.69	2.36	2.38

Source : Sharma *et al.* (2010b)

inter-planted or under sown with a main crop, to serve the purposes of mulch, such as weed suppression and regulation of soil temperature and other environmental benefits. The concept of live mulching is based on mixed cropping whereby fast-growing legume is established before or simultaneously along with widely spaced season grain crops and returned to the soil at an appropriate stage. At Dehradun, Uttarakhand, in sunhemp (*Crotalaria juncea* L.), *dhaincha* (*Sesbania bispinosa* (Jacq.) W. Wight; Syn. *S. aculeate* (Willd.] Pers.) and cowpea [*Vigna unguiculata* (L.) Walp.] as live mulch in maize–wheat cropping system, legume mulch accumulated 1.09–1.17 t/ha dry biomass and added 27.9–31.0 kg N/ha at 30 days after sowing and increased wheat yield by 13.3–14.0% (Sharma *et al.,* 2010a,b; Table 5).

Plastic mulching: Plastic-film mulching plays an important role in agriculture owing to its ability to improve crop yields and WUE by maintaining soil moisture, suppressing weeds and increasing soil temperature. Plastic mulches directly affect the microclimate around the plant by modifying the radiation budget of the surface and decreasing the soil water loss (Liakatas *et al.,* 1986). The colour of plastic-film mulch largely determines its energy-radiating behaviour and its influence on the microclimate around a plant. The efficiency with which black mulch increases soil temperature can be improved by optimizing the condition for transferring heat from the mulch to the soil. To overcome negative environmental problems caused by persistent plastic waste from plastic mulch, biodegradable plastic mulches (BDM) have been developed as a promising alternative, providing a sustainable and environment-friendly solution for agricultural activities. Biodegradable plastic mulch use is on the rise as it provides many of the benefits of PE mulch with the advantage of being tilled in or composted at the end of the season, avoiding the disposal problems of plastic mulch. Liu *et al.* (2021) reported that, BDM treatments increased maize, wheat, cotton and potato (*Solanum tuberosum* L.) yields by 26, 24, 26 and 18%, and water-use efficiency by 24, 23, 15 and 20%, respectively. Beneficial effect of plastic mulch in increasing vegetable yields is given in Table 6.

Table 6. Increase in yield of vegetable crops through plastic mulching

Crop	Yield (t/ha)		Increase in yield (%)
	Un-mulched	mulched	
Broccoli	15.6	25.1	60.7
Cauliflower	18.6	25.0	34.7
Brinjal	36.7	47.0	28.1
Tomato	69.1	94.8	37.3
Okra	6.9	8.6	23.9
Bitter gourd	20.12	25.6	27.4
Chili	16.8	19.7	17.4
cabbage	14.3	19.9	39.1

Source: NCPH, New Delhi **(year?)**

Conservation agriculture

Conservation Agriculture (CA), comprising minimum mechanical soil disturbance and direct seeding, organic mulch cover from residues and cover crops, and crop species diversification through rotations and associations, is now practiced globally on about 125 million ha. The technologies of CA provide opportunities to reduce the cost of production, save water and nutrients, increase yields, increase crop diversification, improve efficient use of resources, reduce runoff and soil loss and benefit the environment (Bhan and Behera, 2014; Rajanna *et al*., 2019). Over the past few years, adoption of zero tillage and CA has expanded to cover about 1.5 million hectares in India (Jat *et al.,* 2012). Araya *et al.* (2011) reported that CA consisting of permanent raised beds with a furrow and bed system, retention of 30% of standing crop residues and zero tillage on the top of the bed reduced runoff by 54%, soil loss by 79% besides increasing wheat yield under rainfed condition in Ethiopia. Similarly, positive influence of CA on soil physical characters and soil organic carbon build up were reported by several authors (Table 7). Sapkota *et al.* (2017) reported that, CT which involves zero tillage direct-seeded rice + zero tillage direct-seeded wheat and residue incorporation increased the soil organic carbon by 6 fold in rice-wheat cropping sequence in Indo-Gangetic Plains.

Table 7. Benefits of conservation agriculture

Conservation agricultural practice	Environmental benefit	Crop yield	Reference
No-tillage with mulch	Increased soil moisture content	9% increase in rainfed wheat	Ziyou Su *et al.* (2007)
No-tillage with crop residue	Reduction in soil loss by 58%	12% increase in maize yield	Rafael *et al.* (2021)
zero till and residue retention	Improved the dry aggregate size distribution	-	Govaerts *et al.* (2009)
Reduced tillage and residue incorporation	Conservation agriculture increased % of water stable aggregates, SOC	Higher soybean yield under CA	Somasundaram *et al.* (2019)
Zero-tillage direct-seeded rice, followed by zero-tillage direct seeded maize + residue retention	Higher SOC content (27%), aggregates, root-mass density	Higher maize yield	Singh *et al.* (2016)
No-till rised bed with residue retention	Higher soil moisture content (17%), maize root mass density	Higher yield of maize	Yadav *et al.* (2018)
Reduced tillage and residue retention	Higher SOC (18%), soil microbial biomass carbon (SMBC), dehydrogenase activity (DHA), and soil NPK	Higher pea (26%) and rapeseed (70%) yield	Das *et al.* (2018)

Cover crops

Cover crops have been defined as crops grown to protect the soil from erosion losses and losses of nutrients via leaching and runoff. Cover crops provide multiple benefits for erosion and runoff control, soil-quality enhancement, nutrient scavenging, and pest suppression. Cover crops reduce sediment production from cropland by intercepting the kinetic energy of rainfall and by reducing the amount and velocity of runoff. These crops increase soil quality by improving biological, chemical and physical properties including: OC content, cation-exchange capacity (CEC), aggregate stability, and water infiltrability. Several researchers have reported the benefits of cover crops in reducing sediment off-site transport. Besides, several studies have reported the impacts of cover crops in increasing nutrient-use efficiencies and C sequestration (Edgar *et al.,* 2009). In an experiment conducted at Nilgiri hills to assess the benefit of winter cover crops viz., buckwheat (*Fagopyrum esculentum* Moench), lupin (*Lupinus albus* L.), Indian mustard [*Brassica juncea* (L.) Czern.)] and fodder oats (*Avena sativa* L.) in potato-based cropping system, Kannan *et al.* (2020) reported that cover crops added N (71.28 kg), P (19 kg) and K (28.5 kg) d under reduced tillage (Table 8).

Vegetative barriers

Vegetative-barrier technology is highly beneficial for marginal and small farmers, since it is cost-effective and easier to establish. In India, different vegetative barriers have been identified for various agro-ecological regions and different soils (Sharda *et al.,* 2006). *Saccharum* spp. for Alfisols of Odisha and Shiwaliks, buffel grass *(Cenchrus ciliaris* L.*)* in dry Vertisols, fountain grass

Table 8. Biomass, nutrition addition and system productivity by different cover crops in potato-carrot sequence

Cover crop	Biomass (t/ha); average of 5 years)		Nutrient addition (kg/ha) from cover crop (average over tillage methods)				System productivity (PEY* t/ha)		SYI	
	CT	RT	C	N	P	K	CT	RT	CT	RT
Fodder oat	4.3	4.5	1980	52.8	17.6	22	59.7	64.1	0.66	0.79
Lupin	2.8	2.2	1064	50	12.5	15	57.7	58.1	0.73	0.66
Buckwheat	3.2	3.8	1155	56	17.5	10.5	62.7	61.3	0.70	0.73
Mustard	4.6	4.9	1615	71.2	19	28.5	63.9	69.0	0.72	0.75
No cover crop							55.6	54.4	0.61	0.63
Average	3.72	3.85					59.9	61.4	0.68	0.74

PEY, Potato-equivalent yield ; SYI, sustainability yield index
Source : Kannan *et al.* (2020)

Table 9. Change in slope (%) due to the vegetative barrier of *Geranium* on sloping land (S_6) and land improvement index (LII)

Original slope	Average slope after		LII after	
	3 years	4 years	3 years	4 years
21.54	17.21	16.77	20.10	22.14
25.59	20.75	20.27	18.91	20.79
28.21	23.69	22.73	16.02	19.43
(Mean) 25.11	20.55	19.92	18.35	20.79

Source: Muralidharan *et al.* (2008)

(*Pennisetum hohenackeri* Hochst. ex steud.*)* in dry Alfisols, *Pennisetum maxicanum* (E. Fourn.) B. D. Jacks] in the sub-humid lower western Himalayas, blue panic grass (*Panicum antidotale* Retz.) and mission grass [*Pennisetum polystachion* (L.) Schult.] for North-Eastern region were identified as effective vegetative barriers. At Dehradun, under rainfed condition, planting 2 rows of grass palmarosa [*Cymbopogon martini* (Roxb) W. Watson] at 1 meter vertical interval in 2% slope reduced runoff, soil loss and increased the soil-moisture availability and yields of maize–wheat cropping system (Gosh *et al.,* 2015). Dass *et al.* (2011) reported that sambuta grass (*Saccharum* spp.) + trench-cum-bund treatment resulted in lesser runoff (8.1%) and soil loss (4.0 t/ha), followed by vetiver grass [*chrysopogon zizaniodes* (L.) Roberty; syn *Vetiveria zizanioides* (L.) Nash] + trench-cum-bund treatment (runoff 9.8%, soil loss 5.5 t/ha) compared to conventional practice in Odisha. Kannan *et al.* (2019) observed that, Hybrid Napier grass [*Cenchrus purpureum* (Schumach.) Morrone] recorded the higher soil-conservation efficiencies besides providing fodder in Nilgiri hills of India. In the field having 25% slope at Nilgiri hills, plating geranium across the slope at 10-m interval revealed lesser soil loss and higher potato-equivalent yield. It was observed that, within 3 years, the original slope of 25.11% was reduced to 20.55% between the barriers resulting in a Land Improvement Index of 18.35 (Muralidharan *et al.,* 2008, Table 9).

Geo-textiles

Geo-textiles are woven nets of fibre made from jute (*Corchorus* sp.), coir (*Cocos nucifera* L.) or any other natural fibre used in soil conservation or any other soil-related constraints in crop production. Several studies have reported the benefits of geo-textiles in river-bank protection and slope stabilization. Adhikari and Shankar (2018) reported that, application of jute geo-textiles increased the groundnut yield by 64.2% and soil organic matter by 53% under rainfed condition in West Bengal. At Dehradun, Uttarakhand, on a 4% land slope, agro-geo-textiles (AGT) prepared from giant-cane (*Arundo donax L.*) placed at 1-m vertical intervals recorded 36% higher maize grain yield (2.8 t/ha) than maize crop raised without AGT (conservation agriculture only). This treatment also reduced runoff (24%) and conserved soil losses (8.22 t/ha/year). Productivities of succeeding pea (*Pisum sativum* L.) and wheat crops were enhanced by 122 and 36%, respectively (Singh *et al.,* 2019). Manivannan *et al.* (2018) reported that, 700 GSM open weave Jute geo textiles (JGT) proved to be more effective in reducing runoff, soil and nutrient loss and increased soil moisture-retention capacity of the soil (Table 10).

SOIL-AND WATER-CONSERVATION MEASURES AT TERRACE LEVEL

Contour bunds and graded bunds

Field bunding across the slope retains run-off in the cultivated field and facilitates its infiltration. Contour bunds are laid across the major land slope along the contour lines in the areas, having 1.5 to 6% land slope and having less than 600 mm annual rainfall. The minimum height of contour bund is 50 cm, with a cross section of 1.61 m^2, having a vertical interval of 0.9 m and the horizontal interval between the bunds may vary from 50 to 70 m, depending on the land slope. The surplus runoff is safely disposed through waste weirs. The graded bunds are constructed with a longitudinal grade of 0.2 to 0.4%, having a vertical interval of 0.75 m to divert the runoff

Table 10. Effect of different open weave jute geo textiles (JGT) on runoff

Year	Rainfall (mm)	Runoff (mm)				Per cent runoff to rainfall			
		500 GSM JGT	600 GSM JGT	700 GSM JGT	Control	500 GSM JGT	600 GSM JGT	700 GSM JGT	Control
2012	798	86.5	58.3	66.6	140.0	10.8	7.3	8.3	17.5
2013	1,142	85.0	69.5	57.1	174.3	7.4	6.1	5.0	15.3
2014	1,098	79.4	49.8	30.4	169.0	7.2	4.5	2.8	15.3
Mean	1,013	83.6	59.2	51.4	161.1	8.5	6.0	5.4	16.0

Source: Manivannan *et al.* (2018)

from the fields. The cross-section area of the bund is 0.83 m^2 and the horizontal distance is 60 to 70 m. These bunds are more suitable for black soils with greater waterlogging in the periods of intense rainfall. In loamy soil of semi-arid region of Rajasthan, field bunding significantly increased mean Indian mustard seed yield by 14.4% and biological yield by 15.3% over no-bunding because of increased availability of soil moisture. Water-use efficiency also increased by 9.7 kg/ha-mm (Regar *et al.,* 2007).

Bench terracing

Bench terracing is a widely practiced soil-conservation measure in hilly areas, having high degree of slopes. It comprises transforming original steep land into series of level strips supported by risers. It breaks the length of slopes and reduces the degree of slopes as well thereby conserving moisture and soil for better crop production (Sharda *et al.,* 2006). Though it is recommended for 16–33% slope, bench terracing is being practiced up to 50% slope in Nilgiri and Himalayan hills owing to socio-economic conditions. Bench terraces may be outward sloping, levelled or inward sloping based on crops grown, rainfall and soil. Levelled or table-top bench terrace is recommended for medium rainfall region with highly permeable deep soil. Inward-sloping bench terracing is more effective in high-rainfall areas for vegetable crops which require good drainage and are susceptible to water stagnation. In high-rainfall regions like Nilgiri hills inward-bench terracing with 2.5% and 1% longitudinal gradient is recommended for safe disposal of water.

Puertorican terraces

Formation of bench terrace by conventional half cut and half fill method is expensive and if proper soil-depth surveys are not conducted, will result in exposure of subsoil leading to reduced crop yields in addition to the area lost under risers which is equal to the per cent slope of land for 1.1 batter. To overcome these undesirable effects, studies on different types of terraces at Udhagamandalam (Ooty) revealed that, Puertorican terrace with vegetative barriers using Guatemala grass (*Tripsacum laxum* Nash) and hybrid napier reduced the cost of construction to one-sixth and one-third of the cost involved in the traditional method (Padmanabhan *et al*., 1988). Mixed vegetative barrier of 2 rows of pineapple [*Ananas comosus* (L.) Merr.] and 1 row of Guatemala grass downstream at 1.0 and 1.5 m vertical interval was also successful in the formation of terraces at Gudalur, Tamil Nadu. This technology is low-cost, easy to adopt, economical and eco-friendly. Puertorican terraces can be developed by first establishing a mechanical (earthen bunds or stone walls) or vegetative (a quick-growing stiff-stemmed vegetation that can stand the pressure of oncoming soil) barrier at the desired vertical along contour or graded line and then moving the soil against the barriers year after year at the time of preparation of cultivation, without incurring any extra expenditure. Studies have indicated that, regular bench terraces can be developed by this method over a period of 3 to 5 years (steeper the slope, shorter is the duration). The recommended soil-moisture measures based on rainfall is given in Table 11.

Table 11. Recommended soil- and moisture conservation measures for different rainfall zones in India

Seasonal rainfall (mm)			
<500	500-750	750-1000	>1000
Contour cultivation	Contour cultivation	BBF (Vertisols)	BBF (Vertisols)
Conservation/ dead furrows	Conservation furrows	Conservation furrows	Field bunds
Ridges and furrows	Ridging	Sowing across slope	Vegetative barriers
Sowing across slope	Sowing across slope	Tillage	Graded bunds
Mulching	Vegetative barriers	Vegetative barriers	Vegetative bunds
Scoops	Scoops	Small basins	Chos
Compartmental bunding	Tied ridges	Vegetative bunds	Level terrace
Graded border strips	Mulching	Field bunds	
Tied ridges	Zing terrace	Graded bunds	
Off-season tillage	Off-season tillage	Nadi	
Inter-row water harvesting system	BBF	Zingg terrace	
Small basins	Inter-row water harvesting system		
Contour bunds	Small basins		
Field bunds	Modified contour bunds		
Khadin	Field bunds		
Graded bunds	Graded bunds		

BBF, Broad bed and furrow

WATER HARVESTING AND RECYCLING FOR CLIMATE-RESILIENT AGRICULTURE

Rainwater management is one of the critical components in rainfed farming, and the successful crop production depends on *in-situ* moisture conservation, surplus runoff-water collection, storage and recycling (Srinivasarao, 2017). Further, the importance of rainwater harvesting for agriculture is now more urgent with increased climatic variability and higher frequency of extreme weather events (IPCC, 2014). Over the recent decades, interventions on rainwater harvesting have been an important component of rural and agricultural development programmes in India, and many water-harvesting structures were created with the public funding from schemes like Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS), Integrated Watershed Management Programme (IWMP), National Horticultural Mission (NHM) and Pradhan Mantri Krishi Sinchai Yojana (PMKSY). The proper design of a water-harvesting system in a high-rainfall region should take into account the spatial and temporal behaviour of rainfall, water requirement of the crops, in addition to catchment characteristics. Srivastava (2001) reported that, for a rice-based cropping system in eastern India, catchment: command ratio of 3.0 and tank size of 1750 m^3/ha command area is required, which facilitates desirable moisture regime for rice and two irrigations to succeeding crop. Kannan *et al.* (2017) found that, in high-rainfall area of the Nilgiri hills, harvesting runoff water in 800 m^3 size was sufficient to take up 1 additional summer crop and supplemental irrigation to the rainy season (*kharif)* and winter season (*rabi)* crops, with cropping intensity increased from 200 to 300% besides increasing farm income to the tune of 111%.

In the rainfed rice-production system of Eastern India, *in-situ* moisture conservation of rainwater through optimum weir height and conservation of run-off water through runoff collection refuge (5–8% of area of individual field) at the end of the field and supplemental irrigation increased the rainfed rice productivity from 1.8 t/ha to 4.53 t/ha and cropping intensity to 176% (Mishra *et al.,* 2003). At watershed level, Kannan *et al*. (2006) evaluated the tank-cum-open dug-well system suitable for plateau region of eastern India. The system comprises of a series of tanks with open dug-wells in the recharge zone of the tank that re-harvest back the seepage water. Thus, the rainwater remaining in the tank as well as partial seeped water is used for providing round the year irrigation.

AGROFORESTRY SYSTEMS FOR RESOURCE CONSERVATION

Agroforestry practices are based on the idea that trees increase nutrient cycling, improve soil fertility and microclimate that support the growth of annual crops (Nair *et al.,* 1999). Agroforestry systems are more profitable and less risky than the traditional agriculture systems because of diversified products and services (Sollen-Norrlin *et al.,* 2020). The root-systems of trees are capable to explore deep soil for water and nutrients, which will help in overcoming the droughts. Additionally, higher litter fall and soil porosity, increased soil cover and reduced runoff (Table 12), leading to increased water infiltration and soil moisture in the soil profile which can reduce moisture stress during less-rainfall years (Hombegowda *et al.,* 2020a). The advantages of AFS over mono agriculture have stimulated renewed interest in the practice of tree cultivation, particularly in areas where they had previously been removed (Beer *et al.,* 1998).

Major advantages of agroforestry systems

Agroforestry systems **(**AFSs) have advantages over conventional agricultural and forest-production systems through increased productivity, economic benefits, sustainability, diversified products, while simultaneously providing numerous other environmental benefits including maintaining soil health and sequestering atmospheric carbon (Kumar, 2011; Nair, 2011). Hence, today AFSs have been recognized as the integrated applied science that has the potential for addressing many of the land-management and environmental problems.

Improvement of soil quality and fertility: It is widely recognized that establishing trees in degraded ecosystems improve soil quality (Pandey, 2010) and enhance soil productivity through biological nitrogen fixation, Soil-carbon sequestration, efficient nutrient cycling, and deep capture of nutrients and water from soils (Nair, 2011). Therefore, growing trees in association with crops can also improve soil quality or reduce the risks of soil degradation (Arunachalam *et al.,* 2002). It was known that the soils under AFS tree canopies were rich in soil organic carbon (SOC) content, moisture availability and nutrient status owing to microclimate moderation (Tscharntke *et al.,* 2011). Hombegowda *et al.* (2015) reported that, home garden and coffee (*Coffea Arabica* L.) agroforestry systems can improve the SOC level up to 85 and 79%, respectively of the forest SOC level in the long term.

Erosion control: The erosion control effect of AFS is most pronounced in steeply sloping land with intense rainstorm events. In hedgerow AFS, the contour bund planted hedgerows significantly reduce the runoff and soil loss while stabilizing the bunds (Hombegowda *et al.,* 2020a). In a long-term field experiment at Eastern Ghats, hedgerows of Gliricidia reduced run-off by 29%, soil loss by 45–48%, and loss of soil organic carbon by 42–47%

Table 12. Agroforestry practices with potential for soil erosion control (Adapted from Young 1989)

Agroforestry practice	Suitable climatic condition	Remarks
Plantation crop combinations	Humid to moist sub-humid climate	Densely planted combinations of plantation crops with multipurpose trees appear to control erosion effectively on at least moderate slopes and improves soil fertility.
Multi-storey tree gardens, including homegardens	Mainly developed in humid and moist sub-humid climates, but possible potential in drier regions	Possess an inherent capacity to control erosion through multi-storey canopy in combination of herbaceous cover with abundant litter.
Hedgerow intercropping (alley cropping) and barrier hedges	Humid, sub-humid and possibly semi-arid climates	A considerable apparent potential to combine erosion control with stable use on gentle to moderate slopes, more speculative potential on steep slopes.
Trees on erosion-control structures	Any	Supplementary use of trees stabilizes earth structures and gives production from land they occupy.
Windbreaks and shelterbelts	Semi-arid zone	Proven potential to reduce wind erosion.
Silvi-pastoral practices	Semi-arid and sub-humid climates, plus some humid	Opportunities for inclusion of trees and shrubs as part of overall programme of pasture improvement.
Reclamation forestry leading to multiple use	Any	Potential for planned design and development.
Combination of the above in integrated watershed management	Any	Substantial opportunities include agroforestry with other major kinds of land use in integrated planning and management.

(Hombegowda *et al.,* 2020a). Similar finding on the significant reduction of run-off and soil loss under different AFS were reported across different locations (Adhikary *et al.,* 2017). Long-term practice of hedgerow cultivation aid in the formation of terraces, especially on the lower parts of slopes (Hombegowda *et al.,* 2020a). Overall, trees in the agricultural lands help in reduction of erosion and soil loss directly by canopy cover effect, through fall and physical barrier in slope, and indirectly by improving soil health, porosity, physical, chemical and biological properties.

Enhancement of crop yield: Crop yield increases in fields adjacent to agroforestry systems have been reported in many studies. The agroforestry system helps in crop production through the microclimate modification, reduced wind speeds, improved soil-moisture availability and improvement of soil health. During drought, deep root-systems of trees showed a positive complementarity with annual crops by exploring a larger soil volume for water and nutrients (Hombegowda *et al.,* 2020b). Furthermore, increased soil porosity, reduced runoff and increased soil cover lead to increased water infiltration and retention in the soil profile and this conserved moisture would help in crop production during dry periods (Siles *et al.,* 2010). Hombegowda *et al.,* (2020a) recorded a 30% increase in the finger millet (*Eleusine coracana* Gaertn.) grain yield under *Gliricidia* hedgerow intercropping system in the sub-humid climate in Eastern Ghats, India. Water-isotope study in coffee agroforestry systems from Western Ghats, India, showed the complementary use of soil water between the coffee plants and mango (*Mangifera indica* L.) and silver oak shade trees during high water requirement season (Hombegowda *et al.,* 2020b). In many other experiments, it is evident that the higher biological yield was recorded under the trees than in the open area (Adhikary *et al.,* 2017). Further, tree-based production systems often produce crops of higher value than mono crops (Tscharntke *et al.,* 2011).

Carbon sequestration: In recent years, AFSs have become popular for their carbon sequestration potential (Table 13). In regard to climate change, the important regulating service offered by AFSs on a global scale is mitigation of atmospheric CO_2 concentration through its sequestration in AFSs (Haile *et al.,* 2008; Nair *et al.,* 2009). Carbon sequestration in AFSs has 2 distinct but related components: sequestration in plant biomass (above-ground and below-ground wood) and in soil as SOC. A study by Roshetko *et al.* (2002) indicated that, C stock stored by Sri Lankan's home garden in their above-ground biomass is equivalent to secondary forests of similar age in the same area. Nair *et al.* (2009) reported that the wood carbon-sequestration potential (CSP) of AFS ranged from 0.29 to 15.21 Mg/ha/year. The CSP is basically linked to the ecosystem productivity of the AFS which in turn depends on tree species composition, stand age, site characteristics and management practices

Table 13. Soil organic carbon sequestration rates in agroforestry systems

Agroforestry systems/species	Location	Age (years)	Soil depth (cm)	Sequestration rate (Mg C/ha/yr)	Reference
Alley cropping system					
Hybrid poplar (*Populus* × *canadensis*) + wheat soybean and maize rotation	Southern Canada	13	0-20	0.30	Oelbermann *et al.* (2006)
Gliricidia sepium + ûnger millet	Eastern Ghats, India	5	0.40	1.8	Hombegowda *et al.* (2020b)
Gliricidia sepium + upland paddy	Eastern Ghats, India	5	0.40	2.19	Adhikary *et al.* (2017)
Erythrina poeppigiana + maize and bean	Costa Rica	19	0–20	1.79	Oelbermann *et al*. (2006)
Other agroforestry systems					
Intercropping systems: hybrid poplar + barley	Southern Canada	13	0–20	1.04	Calculated from Peichl *et al*. (2006)
Multi-strata agroforest: cacao (*Theobroma cacao*) + *Erythrina poeppigiana*	Costa Rica	10	0–45	4.16	Calculated from Beer *et al*. (1990)
Multi-strata agroforest: cacao + *Cordia alliodora*	Costa Rica	10	0-45	1.55	Calculated from Beer *et al*. (1990)

Adapted from Lorenz and Lal, (2018)

Table 14. Summary of soil carbon stock under tropical agroforestry systems

Agro-ecological zone	Agroforestry system	Soil C stock (Mg C/ha)
Humid lowlands	Shaded perennial systems	21–35
	Alley cropping	10–35
	Improved fallow	135–149
	Homegardens	142–156
	Coffee under shade trees	136–169
	Tree intercropping	27–78
	Silvipasture	96–173
	Woodlots	61–75
Tropical highlands	Shaded perennial system	21–97
	Silvipastoral systems	132–173
Arid and semiarid lowlands	Live fences	5.2–24
	Fodder banks	33
	Tree intercropping systems	
	Crop dominated	20–70
	Fodder dominated	25–80
	Fuelwood dominated	25–90
	Shelterbelts	39

Adapted from Nair *et al.,* (2009), Raji and Ogunwole, (2006), Hombegowda *et al.,* (2015)

(Hombegowda *et al.,* 2015). In general, AFS in the low-rainfall regions have relatively lower CSP than the high-rainfall region.

It is reported that AFSs have higher SOC-sequestration potential than pastures or field crops growing under similar ecological conditions (Roshetkot *et al.,* 2002; Nair *et al.,* 2009). The potential of AFSs as C storage varies, depending on the species composition, tree age, climatic condition, soil texture, soil aggregation and its management regimes (Saha *et al.*, 2009). The C-sequestration potential estimates range from 0.29 to 15.21 Mg/ha/year above ground and 30-300 Mg C/ha up to 1m depth in the soils (Table 14; Nair *et al.,* 2010). Due to the current climate change, the role of AFSs in capturing atmospheric

CO_2 and storing the C in plant parts and soil become an important area of research during the recent years.

CONCLUSION

Many studies have clearly indicated that, significant loss in crop productivity due to the extreme-weather events induced runoff and soil erosion. Under this condition, adoption of site-specific agronomic and agroforestry measures enhances the crop performance by improving the soil moisture and controlling the runoff and soil loss from the agriculture lands. In many case studies, biological or vegetative measures are preferred in soil and water conservation, as they are eco-friendly, sustainable and cost-effective in reducing the impact of extreme-weather events. Among the tillage practices, reduced tillage and minimum tillage practices conserve greater rainfall and improve soil physical properties, grain yield and water-use efficiency. However, the effectiveness is influenced by rainfall, soil type and kind of crops. Crops like maize, pigeonpea, cotton, castor, soybean and sunflower (*Helianthus annuus* L.) respond very favourably to deep tillage. For the natural resource conservation, contour cultivation is recommended for all types of soils, rainfall up to 1,000 mm and slope varying from 0.5 to 4%. Among the cultural practices, compartmental bunding, broad bed furrow, ridge and furrow systems, broad bed-furrow system and conservation furrows are effective in controlling erosion rate and in improving the crop performance. Under mulching and residue management, organic, live and plastic mulches and cover-crop management are effective in reducing the kinetic energy of the rain drops, preventing soil erosion, facilitating infiltration, regulating soil-temperature, improving the water-holding capacity of the soil and controlling the weed population. Under slopping lands, vegetative barriers, use of geo-textiles, bench and puertorican terrace are proved effective in reducing the runoff and soil loss. Many agroforestry systems such as alley cropping, windbreaks and shelter belts, multistory gardens and silvi-pastoral practices improve the performance of associated annual crops apart from improving the soil health and soil-carbon sequestration. Many agronomic and agroforestry practices have higher potential to control the runoff and soil loss and improve crop performance during the extreme weather events.

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